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Modification effects of ambient temperature on associations of ambient ozone exposure before and during pregnancy with adverse birth outcomes: A multicity study in China

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ABSTRACT

Background: Epidemiological studies suggest that both ambient ozone (O_3) and temperature were associated with increased risks of adverse birth outcomes. However, very few studies explored their interaction effects, especially for small for gestational age (SGA) and large for gestational age (LGA).

Objectives: To estimate the modification effects of ambient temperature on associations of ambient O_3 exposure before and during pregnancy with preterm birth (PTB), low birth weight (LBW), SGA and LGA based on multicity birth cohorts.

Methods: A total of 56,905 singleton pregnant women from three birth cohorts conducted in Tianjin, Beijing and Maoming, China, were included in the study. Maximum daily 8-h average O_3 concentrations of each pregnant woman from the preconception period to delivery for every day were estimated by matching their home addresses with the Tracking Air Pollution in China (TAP) datasets. We first applied the Cox proportional-hazards regression model to evaluate the city-specific effects of O_3 exposure before and during pregnancy on adverse birth outcomes at different temperature levels with adjustment for potential confounders, and then a meta-analysis across three birth cohorts was conducted to calculate the pooled associations.

Results: In pooled analysis, significant modification effects of ambient temperature on associations of ambient O_3 with PTB, LBW and LGA were observed ($P_{interaction} < 0.05$). For a 10 μg/m^3 increase in ambient O_3 exposure at high temperature level (> 75th percentile), the risk of LBW increased by 28 % (HR: 1.28, 95% CI: 1.13–1.46) during the second trimester and the risk of LGA increased by 116% (HR: 2.16, 95%CI: 1.16–4.00) during the

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entire pregnancy, while the null or weaker association was observed at corresponding low (≤ 25th percentile) and medium (> 25th and ≤ 75th percentile) temperature levels.
Conclusion: This multicity study added new evidence that ambient high temperature may enhance the potential effects of ambient O_3 on adverse birth outcomes.

1. Introduction

Air pollution is one of the top five health risk factors accounting for 6.67 million deaths in 2019 around world (Murray et al., 2020). In recent years, ozone (O_3), formed by the chemical reactions of nitrogen oxides and volatile organic compounds under heat and solar radiation, has become one of the most concerning air pollutants worldwide in addition to particulate matter (PM) (Tang et al., 2021). There is strong evidence that ambient O_3 exposure is associated with increased risks of mortality and various diseases (Kim et al., 2020; Maji et al., 2019; Tian et al., 2021; Vicedo-Cabrera et al., 2020; Zhao et al., 2021). In China, with the rapid development of urbanization and industrialization, ambient O_3 levels are steadily rising and threatening public health (Guan et al., 2021a). Between 2015 and 2020, O_3-related disability-adjusted life years (DALYs) due to all-cause and respiratory deaths in 338 Chinese cities increased by 94.61 % and 96.54 %, respectively (Guan et al., 2021b). Furthermore, the State of Global Air 2020 reported that China had the second highest overall number (93,300) of O_3-related deaths worldwide in 2019 (HEI, 2020).

Adverse birth outcomes, including preterm birth (PTB), low birth weight (LBW), small for gestational age (SGA) and large for gestational age (LGA), can not only cause the disability or death of infants directly, but also increase the risks of chronic diseases, such as diabetes, cardiovascular disease, chronic respiratory disease and neurobehavioral problems in their childhood and adulthood, which may cost a huge economic burden on society (Cristea et al., 2021; Levine et al., 2015; Lewandowski et al., 2020; Trasande et al., 2016; van der Steen and Hokken-Koelega, 2016). Since pregnant women and fetuses are more susceptible to harmful environmental substances, an increasing number of studies have found that exposure to ambient O_3 during pregnancy was associated with increased risks of adverse birth outcomes including PTB, LBW and SGA (Arroyo et al., 2016; Ekland et al., 2021; Huang et al., 2020; Michikawa et al., 2017; Schifano et al., 2013; Wang et al., 2021). For example, a study based on 30 European countries reported that for a 10 μg/m^3 increase in ambient O_3 concentration, the relative risk of PTB increased by 2.7 % [95 % confidence intervals (CI): 0.9 %-4.6 %] (Ekland et al., 2021). In addition to ambient O_3, fluctuation in ambient temperature has also been associated with adverse birth outcomes. Several epidemiological studies suggested that experiencing extreme temperatures, including extreme heat and extreme cold, during pregnancy was associated with increased risks of adverse birth outcomes (Basagaña et al., 2021; Bruckner et al., 2014; Dastoorpoor et al., 2021; Ha et al., 2017; McElroy et al., 2022; Sun et al., 2019; Wang et al., 2020). For instance, one study conducted in Israel reported that exposures to hot (> 90th vs 41st-50th percentile) and cold (≤ 10th vs 41st-50th percentile) temperatures during the entire pregnancy were both associated with a lower mean birth weight (Basagaña et al., 2021).

Apart from chemical composition of O_3 and its precursors, meteorological conditions, such as temperature, humidity, wind and strong sunlight have been considered as the important influencing factors of O_3 formation. Among these meteorological factors, ambient temperature plays the most critical role in O_3 formation due to the temperature dependencies of the hundreds of reactions involved (Dawson et al., 2007). The formation and accumulation of ambient O_3 are usually accompanied by a high temperature (Chen et al., 2019). In terms of the close relationship between ambient O_3 and temperature, and their independent adverse effects on adverse birth outcomes, a hypothesis has arisen regarding whether ambient temperature could modify the associations between ambient O_3 and risks of adverse birth outcomes. Several epidemiological studies have demonstrated the role of ambient temperature in modifying the associations between ambient O_3 pollution and mortality, respiratory and cardiovascular diseases (Fann et al., 2022; Klompmaker et al., 2021; Schwarz et al., 2021; Shi et al., 2020). However, to the best of our knowledge, a few studies have examined the potential modification effects in regard to adverse birth outcomes. The available a few studies have only focused on a few adverse birth outcomes such as PTB and LBW, and the study areas have been restricted to a single city in developed countries (Chen et al., 2018; Sun et al., 2020). Therefore, more studies involving various study cities and adverse birth outcomes are needed to comprehensively explore the modification effects of ambient temperature on the associations between ambient O_3 and adverse birth outcomes.

The aim of this study was to investigate the modification effects of temperature on the associations between ambient O_3 exposure before and during pregnancy and risks of four different adverse birth outcomes including PTB, LBW, SGA and LGA based on a multicity study conducted in three different cities of China.

2. Material and methods

2.1. Study population

This study was comprised of three birth cohorts conducted in different cities in China: Tianjin, Beijing, and Maoming, respectively. Both Tianjin and Beijing are situated in north China with latitudes 38°33″-40°15″N and 39°4″-41°6″N, respectively. Whereas Maoming is situated in south China with latitude 21°22″–22°42″N. The Tianjin birth cohort was based on the project Environmental and LifEstyle Factors iN metabolic health throughout life-course Trajectories (ELEFANT). In this study, we included pregnant women whose last menstrual period and delivery dates were both between June 2014 and June 2016 from Baby ELEFANT (n = 11,092). More details about Baby ELEFANT have been presented in our published article (Chen et al., 2022). Briefly, couples planning a pregnancy filled out pre-pregnancy basic questionnaires voluntarily when they went to designated hospitals in Tianjin for a free preconception health examination, then their birth outcomes were followed up. The Beijing birth cohort was established in Tongzhou Maternal and Child Health Hospital from 2012 to 2018. After extracting the delivery data for mothers-newborn pairs from the routine maternal health clinical records of Tongzhou Maternal and Child Health Hospital, more basic information collected during the first trimester or the first 13 gestational weeks was extracted from the city prenatal health care system (n = 44,015). Detailed information on this birth cohort has been published previously (Wang et al., 2022). The Maoming birth cohort was based on the Maoming Birth Cohort Study (MBCS). Details of the MBCS have been described previously (Xiao et al., 2022). In short, pregnant women who visited Maoming Women and Children's Hospital from October 2015 to December 2018 and agreed to participate in the pregnancy cohort of the MBCS completed basic questionnaires and were followed up on birth outcomes (n = 15,653).

The details of study population inclusions and exclusions in the three birth cohorts are shown in Figure S1-S3 (see Supplemental Material). Specifically, because ambient O_3 concentrations before 2013 was not available in the air pollution datasets used in this study, pregnant women with the date of last menstrual period before March 25, 2013 in the Beijing birth cohort were excluded. In addition, in the Maoming birth cohort, we only included the first delivery records for pregnant women who had given birth to more than one child in the study hospital and excluded the records of second or third delivery, and also excluded the miscoded duplicated pregnant women. Finally, for the Tianjin, Beijing and Maoming birth cohorts, data from 10,916, 36,044, and 9,945 mother-newborn pairs were used in the final analysis, respectively. A number of maternal variables including adverse birth outcomes, ambient pollution and meteorological data, maternal age, ethnicity, education, body mass index (BMI) before pregnancy, occupation, gravidity, parity, season of conception, neonatal birth year, sex and birth weight, a diagnosis of any chronic disease or infectious disease (including hypertension, type 2 diabetes, cardiopathy, chronic kidney disease, epilepsy, hepatitis B, phthisis, sexually transmitted disease, and cancer) were collected in the three birth cohorts, and equally coded in statistical analysis. Maternal and husband smoking, and alcohol drinking statuses were only collected in the Tianjin and Maoming birth cohorts. All three birth cohorts have obtained ethical approvals: The Tianjin birth cohort was approved by the ethical committee of Tianjin Medical University (No.: TMUhMEC2016022); The Beijing birth cohort was approved by the institutional review board of Peking University (No.: IRB00001052-18010); and the Maoming birth cohort was approved by the Human Studies Committee of Sun Yat-sen University (No.: GW2019010).

2.2. Adverse birth outcomes

Four adverse birth outcomes, including PTB, LBW, SGA and LGA were analyzed in this study. Gestational age was determined by the number of completed gestational weeks since the first day of the last menstrual period in the Tianjin and Beijing birth cohorts. For the Maoming birth cohort, gestational age was measured based on the date of the last menstrual period in combination with ultrasound examination. Birthweight was measured within 72 h after the birth of neonate in all three birth cohorts. PTB was defined as the delivery with a gestational age shorter than 37 completed weeks. LBW was defined as neonate's birth weight < 2500 g. SGA and LGA were defined as neonate's birth weight less than the 10th percentile and more than the 90th percentile for the gestational age based on the Chinese birth weight curve in different neonatal sex, respectively (Zhu et al., 2015). All adverse birth outcomes were sorted into dichotomous variables.

2.3. Ambient pollution and meteorological data

In the present study, ambient maximum daily 8-h average (MDA8) O_3 was applied to reflect the levels of daily O_3 exposure. We collected daily MDA8 O_3 and average $PM_{2.5}$ exposure concentrations for each pregnant woman from the preconception period (12 weeks before pregnancy) to delivery by matching their home addresses with the Tracking Air Pollution in China (TAP) datasets, and the spatial resolution of both MDA8 O_3 and $PM_{2.5}$ was 10-km. The TAP O_3-10 km was estimated by establishing a data-fusion algorithm that combines *in situ* observations, satellite remote sensing measurements, and model results from the community multiscale air quality model. Fivefold cross-validation (CV) was used to evaluate the performance of O_3 estimation and the results showed that the estimate was highly correlated with the *in situ* observations of the MDA8 O_3 ($R^2 = 0.70$) (Xue et al., 2020). The TAP $PM_{2.5}$–10 km was estimated by applying a two-stage machine learning model coupled with the synthetic minority oversampling technique and a tree-based gap-filling method based on multi-source data. Three CVs were used to evaluate the performance of $PM_{2.5}$ estimation and the averaged out-of-bag CV R^2 was 0.83 for different years (Geng et al., 2021). A total of 202, 357, and 179 10-km exposure points were applied to the exposure grids in the Tianjin, Beijing and Maoming birth cohorts, respectively. In addition, both pregnant women's daily average temperature (℃) and daily average relative humidity (%) from the preconception period to delivery for three birth cohorts were all collected from the China Meteorological Administration (https://data.cma.cn/) by matching their home addresses using proximity model. In this study, the temperature and relative humidity data derived from the China Meteorological Administration were real-time observations at fixed monitoring sites with an hourly temporal resolution. The numbers of fixed monitoring sites matched for the Tianjin, Beijing and Maoming birth cohorts were 10, 23, and 5, respectively.

Five exposure windows in the present study were explored: 1) the preconception period (12 weeks before pregnancy); 2) the first trimester (gestational weeks 1–13); 3) the second trimester (gestational weeks 14–26); 4) the third trimester (gestational weeks 27 to delivery); and 5) the entire pregnancy. Pregnant women's exposure concentrations of MDA8 O_3, $PM_{2.5}$, temperature and relative humidity during each exposure window were calculated by averaging their daily data during the corresponding exposure window.

2.4. Statistical analysis

A two-stage procedure was performed in the present study to estimate the health effects of ambient O_3 and temperature on adverse birth outcomes. In the first stage, a series of city-specific analyses were performed based on the Cox proportional-hazards regression model for birth cohorts conducted in Tianjin, Beijing, and Maoming, respectively. In regression models, gestational age was treated as the time-to-event variable and adverse birth outcomes were treated as dichotomous health events. We first evaluated separate associations for exposures to ambient O_3 and temperature over five exposure windows (preconception, first trimester, second trimester, third trimester, or the entire pregnancy) on risks of PTB, LBW, SGA, and LGA in three birth cohorts, respectively.

When evaluating the separate associations for ambient O_3, the MDA8 O_3 concentrations were included as a continuous variable, and potential confounders including maternal age (18–25, 26–30, 31–35, 36–40, 41–45, > 45 years), ethnicity (Han, others), education (less than high school, high school, more than high school), BMI before pregnancy ($\leq$ 18.50, 18.51–23.99, $\geq$ 24 kg/m^2), occupation (manual work, no manual work, unemployed), gravidity (primigravida, multigravida), parity (primiparous, multiparous), season of conception (spring: March-May, summer: June-August, autumn: September-November, winter: December-February), neonatal birth year and birth weight, and ambient temperature and relative humidity were adjusted for in regression models. In addition, neonatal sex (male, female) was also adjusted for in the analyses for PTB and LBW. Ambient temperature and relative humidity were adjusted by including natural cubic splines with 3 degrees of freedom (dfs) (Wang et al., 2018). Hazard ratios (HRs) and 95% CIs were reported for a 10 μg/m^3 increase in ambient MDA8 O_3 concentrations.

To estimate the separate effects of ambient temperature, we stratified the temperature into low ($\leq$ 25th percentile), medium (> 25th and $\leq$ 75th percentile) and high (> 75th percentile) levels based on the 25th and 75th percentiles at corresponding exposure window for each birth cohort (Shi et al., 2020), and treated the medium temperature as the reference level in the regression analysis. People living in different climate regions may have different susceptibilities to temperature (Breitner et al., 2014; Shi et al., 2020; Wilson et al., 2017). Therefore, the chosen of three temperature categories was cohort-specific to account for acclimatization of temperature in the three cities at different latitudes. The aforementioned potential confounders were also adjusted for in the stratified analyses, except for replacing the ambient temperature with ambient O_3 concentrations. HRs and 95% CIs were relative to medium temperature level.

Next, to assess the modification effects of temperature on associations between ambient O_3 and adverse birth outcomes, we estimated the associations of exposure to ambient O_3 over five exposure windows with PTB, LBW, SGA, and LGA for each birth cohort by low, medium and high temperature levels. In addition, a likelihood ratio test was applied by comparing models with and without a multiplicative interaction term for ambient O_3 and temperature levels to test whether ambient temperature could modify the potential effects of ambient O_3.

In the second stage, a meta-analysis was conducted to calculate the pooled associations of ambient O_3 and temperature exposures with adverse birth outcomes respectively, and ambient O_3 exposure with adverse birth outcomes at different temperature levels across three birth cohorts. For pooled analysis, potential modification effects were tested using Q statistics comparing city-specific pooled multivariable-adjusted HRs for ambient O_3 at different temperature levels (Wu et al., 2015).

Several sensitivity analyses were performed to test the robustness of the above results. Firstly, we re-run regression models by adding potential confounders including maternal and husband smoking (yes, no), and maternal and husband alcohol drinking (yes, no) into the models for the Tianjin and Maoming birth cohorts, but not for the Beijing birth cohort, as these variables were not collected in the Beijing birth cohort. Then we established a two-pollutant model to test the possible confounding effect of ambient $PM_{2.5}$. We only considered $PM_{2.5}$ in the two-pollutant model because ambient O_3 and $PM_{2.5}$ are the most concerning air pollutants threatening public health, and the TAP did not include other pollutants. Additionally, we re-run all models after excluding pregnant women with a diagnosis of any chronic disease or infectious disease in three birth cohorts.

All statistical analyses were performed using the statistical software SAS 9.4 (SAS Institute, Inc., Cary, NC) and R 4.0.3 (R Development Core Team 2020), and packages 'survival' and 'splines' in R software were applied.

3. Results

Table 1 shows the maternal and fetal characteristics of three different birth cohorts. Among the participants of the three birth cohorts, maternal aged 18–35 years accounted for 89.6% in Tianjin, 93.6% in Beijing and 89.5% in Maoming; BMI before pregnancy between 18.51 and 23.99 kg/m^2 accounted for 68.7% in Tianjin, 53.4% in Beijing and 60.6% in Maoming; female newborns accounted for 49.5% in Tianjin, 48.7 % in Beijing and 45.8% in Maoming; and the birthweight of newborns was 3,432.6 (SD: 338.3) g in Tianjin, 3,391.5 (SD: 453.4) g in Beijing and 3,010.6 (SD: 519.8) g in Maoming. The proportions of PTB, LBW, SGA and LGA were 2.2%, 0.5%, 1.8% and 4.6% in the Tianjin birth cohort; 3.3%, 2.5%, 6.1% and 11.2% in the Beijing birth cohort; and 13.2%, 11.6%, 15.3% and 3.0% in the Maoming birth cohort, respectively.

The distribution and correlation of daily ambient air pollutants and meteorological variables are shown in Table 2. The average concentration of maternal exposure to ambient O_3 during the preconception period was highest in the Tianjin birth cohort (100.7 μg/m^3) and lowest in the Maoming birth cohort (82.9 μg/m^3), whereas during the entire pregnancy, was highest in the Beijing birth cohort (100.4 μg/m^3) and still lowest in the Maoming birth cohort (82.2 μg/m^3). The average levels of maternal exposure to ambient temperature during the preconception and entire pregnancy periods were both highest in the Maoming birth cohort (22.5 ℃ and 23.4 ℃) and lowest in the Beijing birth cohort (11.8 ℃ and 12.3 ℃). In the Tianjin and Beijing birth cohorts, correlations between maternal ambient O_3 and temperature exposure were highly positive, whereas correlations between maternal O_3 and $PM_{2.5}$ exposure almost were highly inverse (almost all $|r| \geq 0.7$, $P < 0.05$). However, in the Maoming birth cohort, the above correlations were only weakly inverse and positive, respectively (all $|r| \leq 0.4$, $P < 0.05$). Moderately inverse correlations (all $0.4 < |r| < 0.7$, $P < 0.05$) were observed between ambient $PM_{2.5}$ and temperature in all three birth cohorts. The spatial distribution of annual average concentrations of ambient O_3 and $PM_{2.5}$ in Tianjin, Beijing and Maoming from 2013 to 2018 are shown in Figure S4 (see Supplemental Material).

Table 3 summarizes multivariable-adjusted hazard ratios of adverse birth outcomes associated with ambient O_3 exposure during five exposure windows. In meta-analyses, pooled analysis across three birth cohorts showed that ambient O_3 exposure during the preconception period

Table 1
Maternal and fetal characteristics in the three birth cohorts.

birth cohort	Tianjin (n = 10,916)	Beijing (n = 36,044)	Maoming (n = 9,945)
Variable	**N (%)/mean (SD)**	**N (%)/mean (SD)**	**N (%)/mean (SD)**
Maternal age (years)			
18–25	1,899 (17.4)	7,106 (19.7)	3,201 (32.2)
26–30	5,274 (48.3)	17,046 (47.3)	3,860 (38.8)
31–35	2,609 (23.9)	9,574 (26.6)	1,842 (18.5)
36–40	962 (8.8)	2,152 (6.0)	824 (8.3)
41–45	152 (1.4)	162 (0.4)	214 (2.2)
>45	20 (0.2)	4 (0.0)	4 (0.0)
Maternal ethnicity			
Han	10,785 (98.8)	33,911 (94.1)	9,822 (98.8)
others	131 (1.2)	2,133 (5.9)	94 (0.9)
NA	0 (0.0)	0 (0.0)	29 (0.3)
Maternal education			
Less than high school	5,489 (50.3)	3,698 (10.3)	3,701 (37.2)
High school	5,275 (48.3)	7,157 (19.9)	2,851 (28.7)
More than high school	152 (1.4)	24,789 (68.8)	3,393 (34.1)
NA	0 (0.0)	400 (1.1)	0 (0.0)
Maternal BMI before pregnancy(kg/m^2)			
≤18.50	853 (7.8)	3,665 (10.2)	2,352 (23.7)
18.51–23.99	7,502 (68.7)	19,240 (53.4)	6,031 (60.6)
≥24	2,561 (23.5)	7,994 (22.2)	1,562 (15.7)
NA	0 (0.0)	5,145 (14.2)	0 (0.0)
Maternal occupation			
Manual work	8,880 (81.3)	1,198 (3.3)	1,585 (15.9)
Non manual work	1,842 (16.9)	22,753 (63.2)	6,174 (62.1)
Unemployed	194 (1.8)	5,847 (16.2)	1,899 (19.1)
Others	0 (0.0)	5,340 (14.8)	72 (0.7)
NA	0 (0.0)	906 (2.5)	215 (2.2)
Maternal gravidity			
Primigravida	5,980 (54.8)	15,578 (43.2)	3,665 (36.9)
Multigravida	4,936 (45.2)	20,464 (56.8)	6,280 (63.1)
NA	0 (0.00)	2 (0.0)	0 (0.0)
Maternal parity			
Primiparity	6,536 (59.9)	21,347 (59.2)	4,748 (47.7)
Multiparity	4,380 (40.1)	14,542 (40.4)	5,194 (52.2)
NA	0 (0.00)	155 (0.4)	3 (0.0)
Maternal smoking			
No	10,862 (99.5)	–	9,789 (98.4)
Yes	54 (0.5)	–	156 (1.6)
Husband smoking			
No	8,644 (79.2)	–	6,266 (63)
Yes	2,272 (20.8)	–	3,679 (37)
Maternal alcohol drinking			
No	10,621 (97.3)	–	9,625 (96.8)
Yes	295 (2.7)	–	320 (3.2)
Husband alcohol drinking			
No	8,613 (79.0)	–	7,078 (71.2)
Yes	2,291 (21.0)	–	2,867 (28.8)
Neonatal sex			
Male	5,510 (50.5)	18,494 (51.3)	5,390 (54.2)
Female	5,406 (49.5)	17,550 (48.7)	4,555 (45.8)
Season of conception			
Spring (March-May)	2,621 (24.0)	8,605 (23.9)	2,841 (28.6)
Summer (June-August)	4,990 (45.7)	8,771 (24.3)	2,446 (24.6)
Autumn (September-November)	1,572 (14.4)	9,146 (25.4)	1,971 (19.8)
Winter (December-February)	1,733 (15.9)	9,522 (26.4)	2,687 (27.0)
Birth year			
2013	–	15 (0.0)	–
2014	–	5,743 (15.9)	–
2015	5,987 (54.8)	4,862 (13.5)	1,049 (10.5)
2016	4,929 (45.2)	8,781 (24.4)	4,859 (48.9)
2017	–	8,617 (23.9)	2,932 (29.5)
2018	–	8,026 (22.3)	1,105 (11.1)
Birth weight (g)	3,432.6 (338.3)	3,391.5 (453.4)	3,010.6 (519.8)
PTB			
No	10,679 (97.8)	34,863 (96.7)	8,630 (86.8)
Yes	237 (2.2)	1,181 (3.3)	1,315 (13.2)
LBW			

(continued on next page)

Table 1 (*continued*)

birth cohort	Tianjin (n = 10,916)	Beijing (n = 36,044)	Maoming (n = 9,945)
No	10,866 (99.5)	35,141 (97.5)	8,796 (88.4)
Yes	50 (0.5)	903 (2.5)	1,149 (11.6)
SGA			
No	10,720 (98.2)	33,865 (93.9)	8,428 (84.7)
Yes	196 (1.8)	2,179 (6.1)	1,517 (15.3)
LGA			
No	10,412 (95.4)	32,018 (88.8)	9,647 (97.0)
Yes	504 (4.6)	4,026 (11.2)	298 (3.0)

N = number; SD = standard deviation; NA: the data were missing; BMI = body mass index; -: the variables were not available.

was significantly and positively associated with increased risks of PTB, LBW, SGA and LGA; and the same associations were also observed during the first trimester for PTB, LBW and SGA ($P < 0.05$). Multivariable-adjusted hazard ratios of adverse birth outcomes associated with ambient temperature over the five exposure windows are shown in Table S1. The results of pooled analysis showed that compared to medium temperature (> 25th and ≤ 75th percentile), both low (≤ 25th percentile) and high (> 75th percentile) temperatures exposure during the entire pregnancy were significantly associated with increased risks of PTB and LGA ($P < 0.05$).

Table 4 shows the multivariable-adjusted hazard ratios of PTB and LBW associated with O_3 exposure at different temperature levels over five exposure windows. In pooled analysis, significant modification effects of ambient temperature on associations between ambient O_3 and risks of PTB and LBW were both observed during the second and third trimesters (all $P_{interaction} < 0.05$), and the strongest positive associations, though not all were significant, were found at high temperature level compared to low or medium temperature level at the presence of modification. For a 10 μg/m^3 increase in ambient O_3 at high temperature level during the second trimester, the risk of LBW increased by 28 % (HR: 1.28, 95% CI: 1.13–1.46), while null and weaker associations were observed at corresponding low and medium temperature levels, respectively.

Table 5 shows the multivariable-adjusted hazard ratios of SGA and LGA associated with ambient O_3 exposure at different temperature levels over five exposure windows. In pooled analysis, a significant modification effect of ambient temperature was observed on the association between ambient O_3 and the risk of LGA during the entire pregnancy ($P_{interaction} < 0.05$). For a 10 μg/m^3 increase in ambient O_3 at high temperature during the entire pregnancy, the risk of LGA increased by 116 % (HR: 2.16, 95% CI: 1.16–4.00). While the null association was observed at other two temperature levels.

In the sensitivity analyses, consistent significant modification effects of ambient temperature on the pooled associations between ambient O_3 exposure and risks of adverse birth outcomes were observed when additional potential confounders including maternal and husband smoking and alcohol drinking statuses were added into the models for the Tianjin and Maoming birth cohorts (see Tables S2-S3 in Supplemental Material). After adjusting for $PM_{2.5}$ exposure, significant modification effects of ambient temperature on the pooled associations between ambient O_3 exposure and risks of PTB and LBW were remained unchanged (see Tables S4 in Supplemental Material). However, the significant modification effect of ambient temperature on the pooled association between ambient O_3 and LGA was observed during the third trimester but not the entire pregnancy (see Tables S5 in Supplemental Material). After excluding pregnant women with a diagnosis of any chronic disease or infectious disease in three birth cohorts (Tianjin birth cohort, n = 50; Beijing birth cohort, n = 88; Maoming birth cohort, n = 284), the exact same significant modification effects of ambient temperature were observed (see Table S6-S7 in Supplemental Material).

Table 2
Distribution and correlation of daily ambient air pollutants and meteorological exposure over the preconception and entire pregnancy periods among all pregnant women in three birth cohorts.

Exposure period	Exposure	Mean	SD	Percentile			O_3	$PM_{2.5}$	Tempe-rature	Relative humidity
				25th	50th	75th				
Tianjin (n = 10,916)										
Preconception [a]	O_3 (μg/m^3)	100.7	36.7	64.7	114.2	131.0	1.00	−0.86	0.93	0.53
	$PM_{2.5}$ (μg/m^3)	77.5	23.3	60.4	67.9	90.7		1.00	−0.76	−0.31
	Temperature (°C)	15.2	9.4	5.6	18.4	23.3			1.00	0.70
	Relative humidity (%)	53.7	8.5	47.3	51.7	60.5				1.00
Entire pregnancy	O_3 (μg/m^3)	89.6	10.7	81.3	87.6	98.6	1.00	−0.56	0.83	0.34
	$PM_{2.5}$ (μg/m^3)	81.6	9.5	74.8	81.8	88.6		1.00	−0.62	−0.06
	Temperature (°C)	12.5	2.6	10.5	11.7	14.9			1.00	0.31
	Relative humidity (%)	56.3	4.0	53.1	55.8	58.8				1.00
Beijing (n = 36,044)										
Preconception [a]	O_3 (μg/m^3)	95.9	41.7	54.8	100.9	132.1	1.00	−0.87	0.90	0.23
	$PM_{2.5}$ (μg/m^3)	87.8	34.4	60.1	74.1	112.5		1.00	−0.75	−0.16
	Temperature (°C)	11.8	10.4	1.3	12.8	22.1			1.00	0.53
	Relative humidity (%)	57.4	12.8	45.2	56.5	69.4				1.00
Entire pregnancy	O_3 (μg/m^3)	100.4	14.7	88.5	100.2	111.4	1.00	−0.80	0.78	0.24
	$PM_{2.5}$ (μg/m^3)	81.6	18.2	65	84.6	93		1.00	−0.45	−0.23
	Temperature (°C)	12.3	3.2	9.3	12	15.4			1.00	0.44
	Relative humidity (%)	57.1	4.4	54	57.1	60.1				1.00
Maoming (n = 9,945)										
Preconception [a]	O_3 (μg/m^3)	82.9	14.1	74.4	80.8	90.2	1.00	0.17	−0.39	0.09
	$PM_{2.5}$ (μg/m^3)	28.5	8.6	22.1	27.3	32.6		1.00	−0.68	−0.73
	Temperature (°C)	22.5	4.7	18.1	22.7	27.2			1.00	0.35
	Relative humidity (%)	82.7	4.7	80.6	83.5	86.1				1.00
Entire pregnancy	O_3 (μg/m^3)	82.2	7.9	76.6	80.3	86.9	1.00	0.15	−0.25	−0.22
	$PM_{2.5}$ (μg/m^3)	26.9	3.4	24.6	26.8	29.3		1.00	−0.37	−0.59
	Temperature (°C)	23.4	1.6	22.0	23.4	24.8			1.00	0.07
	Relative humidity (%)	82.9	2.3	81.0	83.6	84.6				1.00

SD = standard deviation.
[a] Data for the preconception were obtained up to 12 weeks before pregnancy; spearman correlation analysis was used to test correlations between variables, and the P value were all < 0.05.

Table 3
Multivariable-adjusted hazard ratios of adverse birth outcomes associated with a 10 μg/m^3 increase in ambient O_3 exposure during five exposure windows in three birth cohorts.

Birth outcomes	Exposure period	Tianjin *HR* (95% *CI*)	Beijing *HR* (95% *CI*)	Maoming *HR* (95% *CI*)	Meta-analysis [b] *HR* (95% *CI*)	$P_{heterogeneity}$
PTB	Preconception [a]	1.12 (0.94–1.33)	**1.14 (1.04–1.26)**	**1.22 (1.15–1.30)**	**1.19 (1.13–1.25)**	0.392
	First trimester	**1.33 (1.15–1.53)**	**1.25 (1.13–1.38)**	**1.13 (1.07–1.20)**	**1.21 (1.10–1.34)**	0.046
	Second trimester	0.97 (0.83–1.13)	1.02 (0.94–1.12)	1.06 (1.00–1.13)	1.04 (0.99–1.09)	0.511
	Third trimester	0.65 (0.55–0.75)	1.02 (0.94–1.11)	**1.16 (1.09–1.23)**	0.92 (0.66–1.29)	<0.001
	Entire pregnancy	0.50 (0.41–0.61)	**1.68 (1.45–1.94)**	**1.21 (1.09–1.34)**	0.78 (0.33–1.86)	<0.001
LBW	Preconception [a]	1.28 (0.88–1.86)	**1.14 (1.02–1.27)**	**1.19 (1.11–1.27)**	**1.18 (1.11–1.25)**	0.733
	First trimester	1.23 (0.90–1.67)	**1.26 (1.12–1.41)**	**1.08 (1.01–1.15)**	**1.12 (1.06–1.19)**	0.062
	Second trimester	1.00 (0.71–1.43)	0.99 (0.89–1.10)	1.06 (0.99–1.14)	1.04 (0.98–1.10)	0.563
	Third trimester	1.03 (0.70–1.53)	1.01 (0.92–1.11)	**1.22 (1.14–1.30)**	1.10 (0.95–1.29)	0.005
	Entire pregnancy	0.60 (0.38–0.93)	**1.72 (1.45–2.04)**	**1.24 (1.11–1.38)**	0.89 (0.44–1.81)	0.002
SGA	Preconception [a]	1.00 (0.81–1.24)	**1.08 (1.01–1.16)**	**1.07 (1.01–1.14)**	**1.07 (1.02–1.12)**	0.796
	First trimester	**1.19 (1.00–1.42)**	**1.10 (1.02–1.19)**	1.03 (0.98–1.09)	**1.06 (1.02–1.11)**	0.159
	Second trimester	0.73 (0.61–0.87)	0.99 (0.92–1.06)	1.07 (1.00–1.13)	0.93 (0.75–1.15)	<0.001
	Third trimester	0.75 (0.62–0.91)	1.01 (0.95–1.08)	1.01 (0.96–1.08)	0.93 (0.79–1.11)	0.012
	Entire pregnancy	0.62 (0.47–0.81)	**1.26 (1.12–1.43)**	**1.13 (1.03–1.24)**	0.85 (0.47–1.53)	<0.001
LGA	Preconception [a]	**1.17 (1.03–1.33)**	1.02 (0.97–1.08)	1.12 (0.99–1.28)	**1.05 (1.00–1.10)**	0.089
	First trimester	**1.34 (1.21–1.49)**	1.04 (0.99–1.10)	1.09 (0.97–1.23)	1.15 (0.98–1.34)	<0.001
	Second trimester	0.89 (0.79–0.99)	0.97 (0.92–1.01)	1.09 (0.95–1.24)	0.97 (0.93–1.01)	0.075
	Third trimester	0.85 (0.76–0.96)	0.96 (0.92–1.01)	1.07 (0.94–1.22)	0.95 (0.85–1.07)	0.034
	Entire pregnancy	0.60 (0.51–0.70)	1.07 (0.98–1.17)	1.16 (0.95–1.43)	0.83 (0.44–1.59)	<0.001

All models were adjusted for maternal age, ethnicity, educational level, occupation, body mass index before pregnancy, times of gravidity and parity, season of conception, neonatal birth year, and natural cubic splines with 3 df for mean ambient temperature and relative humidity; for PTB and LBW, neonatal sex was also adjusted.
HR = hazard ratio; CI = confidence interval; black bold indicates significantly ($P < 0.05$) and positively estimated effect.
[a] Data for the preconception period were obtained up to 12 weeks before pregnancy.
[b] meta-analysis for pooled multivariable-adjusted hazard ratios in three birth cohorts. If $P_{heterogeneity} < 0.05$, the estimated effects were based on random-effects models, otherwise the estimated effects were based on fixed-effects models.

4. Discussion

This study explored the modification effects of ambient temperature on associations between ambient O_3 exposure around pregnancy and adverse birth outcomes based on three birth cohorts in Tianjin, Beijing, Maoming, China, respectively. To the best of our knowledge, this is the first study to investigate the modification effects of ambient temperature on ozone's impact on multiple birth outcomes (including PTB, LBW, SGA and LGA) across multicity birth cohorts. The results of pooled analysis showed that exposure to ambient O_3 before and during pregnancy was associated with increased risks of adverse birth outcomes, and these associations were modified by temperature. In the presence of modification ($P_{interaction} < 0.05$), the strongest positive associations between ambient O_3 and risks of PTB, LBW, and LGA were found at high temperature level (HRs ranging from 1.18 to 2.24) but not at low or medium temperature level, though these associations were not all statistically significant. The significantly strongest positive associations of ambient O_3 exposure with risks of LBW and LGA were found at high temperature level during the second trimester and entire pregnancy, respectively.

4.1. *Associations of ambient O_3 exposure with risks of adverse birth outcomes*

A majority of studies have reported that exposure to ambient O_3 during pregnancy was associated with increased risks of PTB, LBW and SGA, though the susceptible exposure windows were inconsistent (Ekland et al., 2021; Le et al., 2012; Li et al., 2020; Li et al., 2019). Early pregnancy, including the first trimester and second trimester, was generally found to be the susceptible exposure window (Arroyo et al., 2016; Michikawa et al., 2017; Rappazzo et al., 2021; Wang et al., 2021), while some studies also reported associations for the third trimester, entire pregnancy or all three trimesters (Ju et al., 2021; Wang et al., 2019). Apart from trimester-specific and entire pregnancy, the preconception period may also be a susceptible exposure window. It has been reported that air pollution exposure before pregnancy was associated with elevated levels of intrauterine inflammation and increased risks of adverse birth outcomes including PTB, LBW, SGA and LGA (Chen et al., 2020; Chen et al., 2022; Nachman et al., 2016). In addition, compared with PTB, LBW and SGA, very few studies have estimated the effects of O_3 exposure on the risk of LGA, and only null or slightly inverse associations were observed (Wang et al., 2019). In this study, we took the preconception period and multiple adverse birth outcomes including LGA into consideration, and the pooled estimates suggest that ambient O_3 exposure during the preconception period was associated with increased risks of PTB, LBW, SGA and LGA, and during the first trimester for increased risks of PTB, LBW and SGA. In city-specific analysis, a few significant inverse associations between ambient O_3 exposure and adverse birth outcomes were observed in the Tianjin birth cohort but not in the other two birth cohorts, though random effect *meta*-analysis was used in the pooled analysis to adjust for this heterogeneity. This systematic heterogeneity may be due to the fact that participants in the Tianjin birth cohort were limited to women planning to get pregnant. They tend to be healthier than the general pregnant women population and could obtain more timely medical treatment when anything unusual occurred during the pregnancy than other pregnant women.

It is widely proposed that systemic inflammation and oxidative stress response induced by ambient O_3 exposure were the main potential mechanisms leading to adverse birth outcomes (Klepac et al., 2018; Rappazzo et al., 2021). For PTB, LBW and SGA, it is found that systemic inflammation and oxidative stress response induced by O_3 could impair the placental function and decrease the *trans*-placental oxygen and nutrient transport (Kannan et al., 2006; Slama et al., 2008). For LGA, the mechanism may involve mother's insulin resistance and abnormal glucose tolerance initiated by O_3-associated systemic inflammation and oxidative stress response (Miller et al., 2016; Thomson et al., 2018; Zhong et al., 2016). At present, evidence for the relevant mechanism hypothesis is insufficient, further experimental studies with different effect endpoints of O_3 exposure are needed to investigate the proposed mechanisms for different adverse birth outcomes.

Table 4
Multivariable-adjusted hazard ratios of PTB and LBW associated with a 10 μg/m^3 increase in ambient O_3 exposure at different temperature levels over five exposure windows in three birth cohorts.

Birth outcome	Exposure period	Temperature level	Tianjin HR (95% CI)	$P_{interaction}$	Beijing HR (95% CI)	$P_{interaction}$	Maoming HR (95% CI)	$P_{interaction}$	Meta-analysis[b] HR (95% CI)	$P_{interaction}$[c]
PTB	Preconception[a]	Low	**1.75 (1.17–2.62)**		**1.32 (1.11–1.56)**		**1.24 (1.08–1.43)**		**1.30 (1.17–1.44)**	
		Medium	0.94 (0.79–1.12)	<0.001	1.01 (0.99–1.04)	0.002	**1.29 (1.19–1.40)**	<0.001	1.08 (0.89–1.30)*	0.229
		High	1.22 (0.93–1.58)		1.24 (0.98–1.56)		**1.33 (1.12–1.57)**		**1.28 (1.13–1.45)**	
	First trimester	Low	0.98 (0.78–1.24)		1.18 (1.00–1.39)		**1.12 (1.02–1.24)**		**1.12 (1.03–1.21)**	
		Medium	1.13 (0.99–1.30)	0.001	**1.08 (1.00–1.15)**	0.223	**1.18 (1.09–1.28)**	0.013	**1.12 (1.06–1.17)**	0.264
		High	**1.60 (1.20–2.14)**		**1.86 (1.70–2.02)**		1.08 (0.91–1.27)		**1.48 (1.06–2.06)***	
	Second trimester	Low	1.05 (0.72–1.52)		0.91 (0.78–1.08)		0.78 (0.67–0.90)		0.85 (0.77–0.95)	
		Medium	0.88 (0.76–1.02)	0.490	1.03 (0.97–1.10)	0.478	**1.17 (1.08–1.28)**	<0.001	1.03 (0.88–1.20)*	**0.017**
		High	0.87 (0.65–1.18)		**1.28 (1.06–1.54)**		**1.35 (1.15–1.59)**		1.18 (0.92–1.50)*	
	Third trimester	Low	0.39 (0.27–0.58)		0.39 (0.33–0.46)		0.59 (0.51–0.68)		0.46 (0.34–0.61)*	
		Medium	0.82 (0.73–0.93)	<0.001	**1.08 (1.01–1.17)**	<0.001	**1.30 (1.19–1.42)**	0.001	1.05 (0.81–1.36)*	**<0.001**
		High	**1.65 (1.37–2.00)**		**3.49 (2.88–4.24)**		0.98 (0.88–1.08)		1.77 (0.86–3.66)*	
	Entire pregnancy	Low	0.85 (0.59–1.22)		0.03 (0.02–0.04)		**1.46 (1.21–1.76)**		0.33 (0.03–3.62)*	
		Medium	1.05 (0.72–1.52)	<0.001	**1.67 (1.32–2.13)**	<0.001	1.01 (0.83–1.22)	0.023	1.22 (0.87–1.69)*	0.166
		High	**2.86 (2.19–3.73)**		**32.62 (26.62–39.96)**		**1.55 (1.28–1.88)**		5.25 (0.85–32.57)*	
LBW	Preconception[a]	Low[d]	–		**1.24 (1.02–1.51)**		**1.20 (1.03–1.40)**		**1.22 (1.08–1.37)**	
		Medium	1.49 (1.01–2.20)	0.128	1.01 (0.98–1.04)	0.016	**1.23 (1.13–1.35)**	0.001	1.16 (0.96–1.41)*	0.614
		High	1.30 (0.75–2.26)		1.10 (0.87–1.40)		**1.42 (1.19–1.69)**		**1.30 (1.13–1.49)**	
	First trimester	Low	0.60 (0.29–1.25)		0.99 (0.82–1.20)		1.07 (0.97–1.19)		1.04 (0.95–1.14)	
		Medium	1.02 (0.77–1.34)	0.006	**1.10 (1.02–1.19)**	0.577	**1.12 (1.03–1.22)**	0.010	**1.11 (1.05–1.17)**	0.125
		High	1.90 (1.14–3.19)		**1.65 (1.22–2.23)**		1.12 (0.92–1.35)		**1.44 (1.04–2.00)***	
	Second trimester	Low	1.01 (0.49–2.06)		0.93 (0.78–1.10)		0.85 (0.73–0.99)		0.89 (0.79–0.99)	
		Medium	0.84 (0.62–1.15)	0.360	1.06 (0.99–1.15)	0.664	**1.12 (1.03–1.23)**	0.002	**1.08 (1.02–1.14)**	**<0.001**
		High	1.84 (0.98–3.44)		**1.27 (1.03–1.56)**		**1.26 (1.06–1.49)**		**1.28 (1.13–1.46)**	
	Third trimester	Low	0.43 (0.20–0.89)		0.43 (0.36–0.52)		0.66 (0.60–0.72)		0.52 (0.37–0.73)*	
		Medium	0.88 (0.69–1.13)	0.002	1.07 (0.99–1.17)	0.001	**1.38 (1.25–1.51)**	0.011	1.11 (0.87–1.42)*	**<0.001**
		High	**4.14 (2.74–6.25)**		**2.89 (2.32–3.60)**		0.99 (0.89–1.11)		2.24 (0.96–5.23)*	
	Entire pregnancy	Low	0.30 (0.16–0.60)		0.04 (0.03–0.05)		**1.60 (1.30–1.96)**		0.27 (0.03–2.21)*	
		Medium	1.69 (0.85–3.38)	0.020	**1.42 (1.10–1.83)**	<0.001	1.03 (0.84–1.26)	0.025	**1.19 (1.02–1.39)**	0.140
		High	2.14 (0.62–7.33)		**25.05 (19.80–31.71)**		**1.52 (1.24–1.86)**		4.48 (0.75–26.81)*	

All models were adjusted for maternal age, ethnicity, educational level, occupation, body mass index before pregnancy, times of gravidity and parity, season of conception, neonatal birth year and sex, and natural cubic splines with 3 df for mean relative humidity.
HR = hazard ratio; CI = confidence interval; black bold indicates significantly ($P < 0.05$) and positively estimated effect.
[a] Data for the preconception period were obtained up to 12 weeks before pregnancy.
[b] meta-analysis for pooled multivariable-adjusted hazard ratios in three birth cohorts. If $P_{heterogeneity} < 0.05$, the estimated effects were based on random-effects models, otherwise the estimated effects were based on fixed-effects models, * $P_{heterogeneity} < 0.05$.
[c] $P_{\text{interaction}}$ for *meta*-analysis were calculated using Q statistic comparing city-specific pooled multivariable-adjusted HRs for ambient O_3 at different temperature levels.
[d] The number of PTB at low temperature level in Tianjin birth cohort was not enough to calculate.

Table 5
Multivariable-adjusted hazard ratios of SGA and LGA associated with a 10 μg/m^3 increase ambient O_3 exposure at different temperature levels over five exposure windows in three birth cohorts.

Birth outcome	Exposure period	Temperatu-re level	Tianjin *HR (95% CI)*	$P_{interaction}$	Beijing *HR (95% CI)*	$P_{interaction}$	Maoming *HR (95% CI)*	$P_{interaction}$	Meta-analysis[b] *HR (95% CI)*	$P_{interaction}$[c]
SGA	Preconception[a]	Low	0.41 (0.17–0.96)		1.10 (0.97–1.26)		**1.25 (1.09–1.43)**		0.98 (0.62–1.55)*	
		Medium	1.14 (0.94–1.38)	0.003	0.98 (0.94–1.03)	0.003	1.02 (0.93–1.11)	0.741	0.99 (0.96–1.03)	0.442
		High	**1.49 (1.10–2.03)**		1.00 (0.86–1.16)		1.04 (0.90–1.21)		1.06 (0.96–1.17)	
	First trimester	Low	0.83 (0.59–1.17)		1.04 (0.94–1.15)		1.05 (0.96–1.16)		1.04 (0.97–1.11)	
		Medium	0.94 (0.80–1.11)	<0.001	1.00 (0.95–1.05)	0.340	0.97 (0.90–1.05)	0.121	0.99 (0.95–1.03)	0.245
		High	**1.43 (1.08–1.88)**		0.95 (0.81–1.12)		1.16 (0.98–1.37)		1.14 (0.92–1.42)*	
	Second trimester	Low	0.34 (0.23–0.51)		0.94 (0.84–1.06)		1.01 (0.88–1.14)		0.70 (0.36–1.37)*	
		Medium	0.80 (0.69–0.93)	<0.001	1.05 (1.00–1.10)	0.992	**1.09 (1.00–1.18)**	0.178	0.98 (0.82–1.17)*	0.303
		High	**2.76 (1.74–4.39)**		1.06 (0.93–1.21)		1.05 (0.91–1.21)		1.40 (0.78–2.52)*	
	Third trimester	Low	0.33 (0.20–0.55)		0.83 (0.73–0.94)		0.90 (0.79–1.03)		0.66 (0.37–1.17)*	
		Medium	1.03 (0.93–1.14)	<0.001	**1.06 (1.02–1.11)**	0.261	**1.09 (1.00–1.18)**	0.778	**1.06 (1.02–1.10)**	0.238
		High	**1.47 (1.09–1.97)**		**1.11 (1.02–1.21)**		0.96 (0.86–1.07)		1.12 (0.91–1.38)*	
	Entire pregnancy	Low	0.50 (0.27–0.93)		0.13 (0.10–0.18)		1.10 (0.90–1.34)		0.41 (0.12–1.44)*	
		Medium	0.71 (0.52–0.97)	0.022	1.12 (0.98–1.28)	0.002	1.10 (0.96–1.27)	0.651	0.99 (0.77–1.27)*	0.112
		High	1.49 (0.74–2.99)		**5.80 (4.35–7.72)**		1.19 (0.99–1.43)		2.20 (0.81–5.98)*	
LGA	Preconception[a]	Low	1.04 (0.65–1.65)		0.98 (0.89–1.08)		1.19 (0.87–1.63)		1.00 (0.91–1.09)	
		Medium	1.11 (0.97–1.26)	<0.001	0.99 (0.95–1.02)	0.441	**1.20 (1.02–1.42)**	0.068	1.07 (0.96–1.20)*	0.137
		High	**1.31 (1.09–1.57)**		1.10 (0.98–1.24)		0.97 (0.70–1.34)		**1.14 (1.04–1.25)**	
	First trimester	Low	0.90 (0.75–1.07)		1.01 (0.94–1.09)		1.08 (0.85–1.37)		1.00 (0.94–1.07)	
		Medium	**1.15 (1.05–1.28)**	<0.001	1.01 (0.98–1.05)	0.985	0.99 (0.85–1.16)	0.886	1.05 (0.96–1.15)*	0.575
		High	1.17 (0.96–1.43)		1.01 (0.91–1.13)		1.07 (0.79–1.46)		1.05 (0.96–1.15)	
	Second trimester	Low	0.86 (0.68–1.11)		1.00 (0.92–1.09)		0.91 (0.66–1.27)		0.98 (0.91–1.06)	
		Medium	0.88 (0.80–0.98)	0.013	1.02 (0.98–1.06)	0.908	1.18 (0.98–1.41)	0.439	1.01 (0.87–1.17)*	0.404
		High	0.98 (0.77–1.26)		1.06 (0.96–1.16)		1.22 (0.88–1.70)		1.06 (0.97–1.15)	
	Third trimester	Low	0.51 (0.38–0.68)		0.82 (0.75–0.90)		1.00 (0.78–1.28)		0.76 (0.52–1.09)*	
		Medium	0.96 (0.90–1.03)	<0.001	1.01 (0.98–1.05)	0.329	1.15 (0.96–1.38)	0.613	1.00 (0.97–1.03)	0.153
		High	**1.53 (1.32–1.77)**		**1.12 (1.04–1.20)**		0.95 (0.73–1.23)		1.19 (0.91–1.55)*	
	Entire pregnancy	Low	0.92 (0.68–1.24)		0.24 (0.18–0.30)		1.42 (0.88–2.29)		0.67 (0.23–1.92)*	
		Medium	0.79 (0.64–0.98)	<0.001	0.98 (0.89–1.09)	<0.001	1.17 (0.85–1.59)	0.869	0.96 (0.88–1.05)	**0.031**
		High	**2.08 (1.63–2.64)**		**3.66 (3.40–3.94)**		1.22 (0.81–1.85)		**2.16 (1.16–4.00)***	

All models were adjusted for maternal age, ethnicity, educational level, occupation, body mass index before pregnancy, times of gravidity and parity, season of conception, neonatal birth year, and natural cubic splines with 3 df for mean relative humidity.
HR = hazard ratio; CI = confidence interval; black bold indicates significantly ($P < 0.05$) and positively estimated effect.
[a] Data for the preconception period were obtained up to 12 weeks before pregnancy.
[b] meta-analysis for pooled multivariable-adjusted hazard ratios in three birth cohorts. If $P_{heterogeneity} < 0.05$, the estimated effects were based on random-effects models, otherwise the estimated effects were based on fixed-effects models, * $P_{heterogeneity} < 0.05$.
[c] $P_{\text{interaction}}$ for *meta*-analysis were calculated using Q statistic comparing city-specific pooled multivariable-adjusted HRs for ambient O_3 at different temperature levels.

4.2. 4.2. Associations of ambient temperature with risks of adverse birth outcomes

Our results of associations between ambient temperature and risks of PTB, LBW, SGA and LGA suggest that low ($\leq 25^{th}$ percentile) or/and high ($> 75^{th}$ percentile) temperatures exposure over the entire pregnancy may increase risks of adverse birth outcomes in comparison with medium temperature ($> 25^{th}$ and $\leq 75^{th}$ percentile). These findings are consistent with several published studies (Basagaña et al., 2021; Basu et al., 2018; Kloog et al., 2018; Sun et al., 2019; Wang et al., 2020). For example, a cohort study based on a national birth cohort study in mainland China reported that both relatively low ($< 5^{th}$ percentile, 9.1 ℃) and high ($> 95^{th}$ percentile, 23.0 ℃) temperatures were associated with an increased risk of PTB in comparison with the threshold (12 ℃) (Wang et al., 2020).

The exact mechanisms for the impact of low and high temperatures on adverse birth outcomes are uncertain. It is biologically plausible that extreme temperatures would induce systemic inflammation and oxidative stress response and damage placental function (Ferguson et al., 2018; Ganesan et al., 2017; Kahle et al., 2015). In addition, for PTB, LBW and SGA, it is proposed that extreme temperatures may also be directly involved in alteration of uterine blood flow and blood viscosity (Dadvand et al., 2011). Heat stress is associated with maternal dehydration which could decrease uterine blood flow and result in insufficient fetal nutrition (He et al., 2016). Furthermore, low temperature leads veins and arteries to narrow and may increase the risk of hypertensive disorders of pregnancy, which might change uteroplacental perfusion and adversely affect fetal growth (Bruckner et al., 2014). For LGA, it may be associated with proliferation of brown adipose tissue induced by low and high temperature. Brown adipose tissue is used for non-shivering thermogenesis, and its development and activation reduce adiposity through higher energy expenditure (Dionicio López et al., 2022).

4.3. Modification effects of ambient O_3 and temperature on adverse birth outcomes

Our results of pooled analysis showed that ambient temperature, especially high temperature could modify the potential effects of ambient O_3 on risks of adverse birth outcomes. The strongest positive associations between ambient O_3 and risks of PTB, LBW, and SGA were observed at high temperature level but not at low or medium temperature levels at the presence of modification. A previous study conducted in California, the USA has also reported similar results. This study estimated the joint effects of heatwaves and ambient O_3 on the risk of PTB, and found that heatwaves could modify the association between ambient O_3 and the risk of PTB (Sun et al., 2020). Our results provide more credible evidence based on a multicity study setting and estimated the potential modification effects for multiple adverse birth outcomes especially for LBW and LGA. This study included three different cities. Among these cities, Tianjin and Beijing, both in north China, have temperate monsoon climates with winters cold and dry. While Maoming, in south China, has a subtropical oceanic monsoon climate with high temperature and humidity over most time of the year. It has been reported that both high temperature and low humidity are conducive to O_3 formation. (Dong et al., 2020). In this study, a positive correlation between maternal ambient O_3 and temperature exposure was observed in the Tianjin and Beijing birth cohorts but not in the Maoming birth cohort. This may be due to the high humidity weather in Maoming. However, we also observed that high temperature enhanced the potential effect of ambient O_3 on risks of adverse birth outcomes in city-specific analysis for the Maoming birth cohort. This finding suggests that the modification effects of high temperature on ambient O_3 are not affected by the climate region.

When estimating the independent and joint effects of ambient O_3 on risks of adverse birth outcomes, a few significant and inverse associations were observed, especially at low temperature. The same associations also were reported in some previous studies (Ha et al., 2014; Sarizadeh et al., 2020). In fact, the health effects of O_3 are double-sided. Except for harmful effects, O_3 can also activate the antioxidant system, potentiate the immune system's response by increasing the production of interferon, and decrease tissue hypoxia (Elvis and Ekta, 2011). It has been reported that exposure to O_3 at non-toxic doses has protective effects against cardiovascular complications, liver injuries, renal injuries, diabetes and its related complications, and harmful effects of UV radiation (Ajamieh et al., 2002; Braidy et al., 2018; Delgado-Roche et al., 2013; Gul et al., 2012; Gultekin et al., 2013). In general, cold temperature is accompanied by low O_3 concentrations. This may help explain the inverse associations of O_3 with adverse birth outcomes observed at low temperature in this study.

4.4. Strengths and limitations

The main strength of our study is that we estimated the modification effects of ambient temperature on associations between ambient O_3 and risks of multiple adverse outcomes including PTB, LBW, SGA and LGA based on multicity birth cohorts. To the best of our knowledge, very few studies have estimated the modification effects of ambient temperature on associations between ambient O_3 and risks of adverse birth outcomes, and no study has focused on adverse birth outcomes for SGA and LGA, let alone include the above four adverse birth outcomes simultaneously in a single study. In addition, in our study, three different cities covering the temperate and tropical zones in China were included, which may help provide more convincing evidence.

There are some limitations in the present study. Firstly, we estimated the exposure data for air pollutants and meteorological factors by matching participants' home addresses. This may not completely reflect the real individual exposure since we did not consider their indoor exposure, outdoor activities and some intervention behaviors, such as the use of air purifiers and air conditioners. Secondly, we did not adjust for gestational diabetes in this study because the data was not collected, though a latest systematic review reported that gestational diabetes was associated with increased risks of PTB and LGA (Ye et al., 2022). However, we have adjusted for pregnancy BMI which was reported to be positively associated with an increased risk of gestational diabetes (Najafi et al., 2019). Thirdly, we did not take other risk factors for adverse birth outcomes, such as socioeconomic status, family size and maternal nutritional status, into consideration due to the unavailability of the data. Therefore, more related multicity studies are needed to further validate our findings.

5. Conclusion

This multicity study added new evidence to the modification effects of ambient temperature on the associations between ambient O_3 and risks of multiple adverse birth outcomes based on three birth cohorts conducted in Tianjin, Beijing and Maoming, China. We found that ambient O_3 exposure around pregnancy was associated with increased risks of PTB, LBW, SGA and LGA, and high ambient temperature may enhance the potential effects of ambient O_3 on risks of PTB, LBW and LGA. These results have potential important public health significance, suggesting that more attention should be paid to the higher risks of adverse birth outcomes in association with ambient O_3 under high ambient temperature. However, it also calls for more future multicity studies to confirm our findings and explore the underlying mechanisms.

CRediT authorship contribution statement

Juan Chen: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Liqiong Guo:** Funding acquisition, Validation, Writing – review & editing, Resources, Supervision. **Huimeng Liu:** Validation, Writing – review & editing. **Lei Jin:** Data curation, Resources, Writing – review &

editing. **Wenying Meng:** Data curation, Resources, Investigation. **Junkai Fang:** Data curation. **Lei Zhao:** Writing – original draft. **Xiao-Wen Zeng:** Data curation, Resources, Investigation. **Bo-Yi Yang:** Data curation, Resources. **Qi Wang:** Project administration. **Xinbiao Guo:** Writing – review & editing. **Furong Deng:** Writing – review & editing. **Guang-Hui Dong:** Project administration, Resources. **Xuejun Shang:** Funding acquisition, Project administration. **Shaowei Wu:** Funding acquisition, Supervision, Conceptualization, Validation, Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2023.107791.

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